Hriday Sainathuni

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EDUCATION

University of Virginia, School of Engineering and Applied Science Charlottesville, VA
Major: B.S. Computer Science **Expected Graduation:** May 2027
Relevant Coursework: Data Structures & Algorithms II, Computer Systems I–II, Discrete Math, Software Development, Cybersecurity
Thomas Jefferson High School for Science and Technology - Prestigious STEM-focused high school **Graduated 2024**

EXPERIENCE

Machine Learning Engineering Intern | *Workday* May 2026 - Aug 2026

- Built cross-session memory for Workday's LangGraph Planning Agent with two custom deepagents middlewares, letting users resume prior sessions instead of rebuilding context
- Engineered a Postgres retriever that ranks prior threads by semantic relevance and recency to inject the top 2 into agent context, achieving 3ms writes and 1.5s recall
- Added a summarization middleware that compresses history past a token budget to prevent context-window overflow
- Built an LLM-as-judge evaluation harness (Claude Opus) over LangSmith traces, scoring 963 confidence-sampled responses with Wilson confidence intervals and significance testing
- Engineered a resumable, memory-bounded pipeline processing 6,392 live agent turns (189K spans) on Kubernetes, using cursor-based extraction and atomic checkpointing to survive pod evictions and surface a 71% failure hotspot
- Fixed 5 recall-pipeline bugs, from session-memory pollution to hallucinated agent suggestions, adding regression tests to prevent recurrence

AI Developer Intern | *Insurity* Dec 2025 - Feb 2026

- Developed full-stack features across the Angular and .NET stack for Insurity's Billing-as-a-Service platform, delivering front-end and back-end work across multiple Jira-tracked tickets
- Built RAG (retrieval-augmented generation) pipelines over policy documents and billing statements: document chunking, OpenAI embeddings, and semantic retrieval indexed in Azure AI Search to ground LLM responses in enterprise content

Software Engineering Intern | *NuAxis (Solutions Engineering Group)* Aug 2025 - Dec 2025

- Built an LLM chatbot on Azure for citizen-facing permit applications that answers requirement questions, guides users through the process, and autofills basic fields from chat input, using OpenAI APIs, LangChain, and PostgreSQL; demoed to a U.S. Fish and Wildlife Service representative
- Built full-stack prototypes with React, MongoDB, and PostgreSQL, and ran content and accessibility analysis across federal websites to improve usability and compliance

AI/ML Intern | *Verizon (Cyber AI Services Team)* May 2025 - Aug 2025

- Built a Dockerized FastAPI REST platform for speech-to-speech GenAI evaluation using Twilio and agentic LLMs, automating adversarial testing across speech modalities
- Integrated Vertex AI into CI/CD and MLOps workflows to automate GenAI vulnerability detection, including prompt injection and robustness testing
- Developed and deployed a GCP-based deepfake synthesis microservice (Chatterbox-TTS, LatentSync); presented threat demonstrations to security and executive teams to raise awareness

Undergraduate Research Assistant | *CRAL Lab at the University of Virginia* Aug 2024 - Aug 2025

- Co-authored an ICRA 2025 paper on decentralized multi-agent coordination for robotic navigation
- Built SMGLib, a multi-agent simulation library using Gym and PyTorch, with PPO reinforcement-learning policies

Research Intern | *Virginia Tech Sanghani Center* Nov 2022 - May 2024

- Developed a GenAI image-classification pipeline using COCO and Conceptual Captions labels to analyze VLM representations
- Contributed to a CVPR 2024 publication on black-box analysis and cross-modal reasoning in vision-language models

SKILLS

Languages: Python, TypeScript, C++, Java, SQL, JavaScript
AI/ML: LangGraph, LangChain, deepagents, LangSmith, embeddings, RAG, OpenAI APIs, PyTorch, TensorFlow, Hugging Face, Vertex AI
Systems & Cloud: PostgreSQL, Azure AI Search, Docker, Kubernetes, AWS (EC2), GCP, Azure, FastAPI, REST APIs, asyncio, pytest, Git
Frameworks & Tools: React, Angular, .NET, ROS